Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/21/2007
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Mechanisms by which Selenomethionine Protects Against Space Radiation Biological Effects		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	akennedy@mail.med.upenn.edu	**Fax:**	FY 215-898-0090
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-0079
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	Oncology Research Division		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/19/2003	**End Date:**	09/30/2007
No. of Post Docs:	6	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per PI (5/07)		
Key Personnel Changes/Previous PI:	James Davis, Ph.D. - replaced Jelena Stewart, Ph.D. during this past year of grant work.		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1517		
Performance Goal No.:			
Performance Goal Text:			
	This research program is focused on the mechanism(s) by which L-selenomethionine reduces the levels of biological effects associated with the exposure of cells to high-energy, highly charged particles known as HZE particles from the new NASA BAF facility at the Brookhaven National Laboratory. It is known that L-selenomethionine can reduce the levels of oxidative stress produced by cellular irradiation with x-rays, protons and HZE particles. It is expected that L-selenomethionine will serve as an effective countermeasure for other types of space radiation induced biological effects, such as the induction of cancer. The studies being performed as part of this grant are exploring the mechanisms by which L-selenomethionine can serve as a countermeasure for radiation effects, including the mechanisms by which it may regulate gene expression related to radiation induced carcinogenesis. This research project is driven by our hypothesis that radiation is capable of initiating an ongoing cellular process (by a high frequency, epigenetic event) that		

Task Description:	is characterized by a state of genetic instability; a later event or events, presumably genetic, then lead directly to malignant transformation. The assumptions are that the heritable epigenetic event is a change in gene expression which is induced by the radiation exposure and that L-selenomethionine is capable of preventing, suppressing or reversing this change in gene expression. Microarray technology is being used to assess the effects of SeM on HZE particle radiation-induced changes in gene expression. Studies on malignant transformation in vitro and gene expression and function utilize HTori-3 cells, which represent a human thyroid epithelial cell model system. Studies with x-rays are also being performed so that Relative Biolgoical Effectiveness (RBE) values for the HZE particles can be determined for radiation induced malignant transformation. By studying the mechanisms by which selenomethionine protects against space radiation induced biological effects, information will be obtained relevant to the mechanisms involved in HZE particle induced radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of the mechanisms by which selenomethionine serves as a countermeasure for radiation induced biological effects could help to develop more effective ways to prevent radiation induced biological damage leading to the known biological effects of radiation (e.g., cell and organ toxicity, mutation and malignant transformation). Research on the mechanisms involved in the prevention of radiation induced biological effects will also have earth benefits, as life on Earth involves continual exposure to various types of radiations which are capable of producing adverse biological effects.
	There has been progress in all of the major areas of research covered by this grant; the progress in each area is summarized as follows. 1) L-selenomethionine (SeM) affects HZE particle radiation induced transformation in vitro. Several publications from this grant work provide data indicating that SeM can prevent HZE particle radiation induced malignant transformation of HTori-3 cells, as measured by the growth of irradiated cells under anchorage independent conditions (i.e., growth in soft agar). 2) Relative Biological Effectiveness (RBE). The results of studies designed to determine the RBEs for HZE particle radiation to produce adverse biological effects have shown that HZE particle radiation is considerably more effective than low linear energy transfer (LET) radiation in the production of both cytotoxicity and malignant transformation in vitro. Results in cytotoxicity studies indicate RBE values of 2.5 and 2.6 for 1-Gev/n and 5-GeV/n iron ion radiation, compared to x-rays, in the clonogenic survival assay using HTori-3 cells. The results of studies performed to determine the RBE values for the induction of anchorage independent growth of HTori-3 cells for iron ions, compared to gamma radiation, is approximately 3. 3) Tumorigenicity Experiments. Experiments were performed with colonies of cells having the ability to grow in soft agar after cellular exposure to a dose of 200 cGy of iron ion radiation. Cells from these colonies of cells were tumorigenic in athymic nude mice. 4) Gene Expression Studies. Our studies on gene expression changes occurring from HZE particle radiation, with and without SeM, were performed at three different time points, which were 2 hours, 6 hours and 16 hours post-irradiation. The results for the microarray studies indicated somewhat different results at these different time periods, related both to the HZE particle radiation effects and to the effects of SeM on the HZE particle radiation induced changes. All of the data clearly show that SeM has major effects on HZE particle radiation-induced differential gene expression. A. 6 hr. time point. For both radiation doses used, a non-toxic (10 cGy) and a slightly toxic (20 cGy) dose of radiation from iron ions, supplementation with SeM leads to a clear change in the pattern of HZE particle radiation-induced differential gene expression. In HTori-3 cells exposed to a 10 cGy radiation dose from iron ions, 218 genes are up-regulated and 82 genes are down-regulated in the absence of SeM; in cells irradiated with the same dose of radiation in the presence of SeM, only 75 out of 218 up-regulated and 15 out of 82 down-regulated genes are still differentially expressed. In HTori-3 cells exposed to a 20 cGy dose of radiation from iron ions, 176 genes are up-regulated and 124 genes are down-regulated in the absence of SeM; in cells exposed to the same dose of radiation in the presence of SeM, only 53 out of 176 up-regulated and 9 out of 124 down-regulated genes are still differentially expressed. Further, a large number of genes were differentially expressed in HTori-3 cells irradiated in the presence of SeM whose expression was not differentially regulated when cells were irradiated, for both radiation doses, in the absence of SeM. In studies using quantitative PCR (qPCR), 17 genes of interest have been partially or totally confirmed to be regulated as the microarray analyses suggested. For the most part, genes involved in the regulation of growth and differentiation have been the candidates found to be regulated by radiation and/or SeM using both microarray and qPCR analyses. These include GADD45alpha, Early Growth Response 1 (EGR1), Cellular Protein p300 (EP300), Transforming Growth Factor Beta (TGFBeta), Bone Morphogenetic Protein 2 (BMP2) and Stratifin (SFN). In addition to these, Selenoprotein W has been shown to be up-regulated with SeM treatment in non-irradiated as well as in irradiated cells. This protein plays an important role in redox-related processes that protect cells from radiation-induced oxidative damage. The significance of GADD45alpha induction by HZE particle radiation exposure with SeM is that the gene product is involved in holding cells up in the cell cycle, so that cell repair processes can occur before cells undergo mitosis and fix the damage produced by the radiation exposure. To determine whether irradiated cells in fact are held up in the cell cycle by SeM treatment, fluorescence activated cell sorter (FACS) analyses were performed. These studies indicated that cells irradiated in the presence of SeM are held up in the cell cycle in the G2 phase so that the transition into mitosis does not occur in a normal fashion. This allows cells to have an increased length of time in this portion of the cell cycle to repair the HZE particle radiation induced cellular and molecular damage in a more complete fashion. The effects of SeM resulting in more time available for cells to repair damage before mitosis occurs suggests that SeM may prevent the malignant transformation of cells by giving the cells more time to repair HZE particle radiation induced damage before it becomes fixed into a form that can not be easily repaired. Since our microarray analyses revealed involvement of several genes associated with cell cycle checkpoint upon irradiation, it seemed reasonable to assume that p53, a major player in regulating checkpoint targets, would be activated as well. We performed immunocytochemistry studies to detect activated p53 proteins in the presence and absence of

Task Progress:	SeM. Increased staining for active p53 (detected by a fluorescently labeled monoclonal antibody against wild-type phosphorylated p53, Ab1620) was increased in cells pretreated with SeM when irradiated with 20 cGy iron ions, as compared to either SeM alone or radiation alone. Similar results were observed utilizing a 10 cGy dose of iron ions. These results are somewhat surprising, in that doses of 10 and 20 cGy of x or gamma radiation can lead to the activation of p53 in other cell types. In HTori-3 cells, our results indicated that 10-20 cGy doses of iron ion radiation did not lead to the activation of p53 by themselves, but when cells are irradiated in the presence of SeM, p53 is activated at these low doses of radiation. It is known that p53 plays a major role in controlling the cell cycle and many other phenomena related to cell growth and malignant transformation. It is hypothesized that many of the gene expression changes observed in irradiated cells treated with SeM are related to effects mediated by p53. The activation of p53, and its associated other gene expression changes, could result in a cell cycle delay which affords cells the time to recover from HZE-particle radiation induced damage prior to entering mitosis. B. 2 hour time point. In the last year of grant work, there was an extensive characterization of the changes in gene expression patterns produced by HZE particle irradiation and the effects of L-selenomethionine used as a countermeasure for the changes in gene expression patterns at 2 hours post-irradiation. HTori-3 cells, cultured in the absence or presence of SeM, were exposed to a low dose (10 or 20 cGy) of iron ions and immediately harvested 2 hours after radiation exposure. Designating changes in gene expression at levels of two-fold or more, approximately 100-200 genes were found to be differentially regulated after exposure to both low doses of iron ion radiation; however, only a few of these genes, which included IL8 and TGFBeta, were affected by treatment with SeM. The SeM effect on the radiation induced differential gene expression patterns for IL8 and TGFBeta was confirmed by qPCR and it was found that mRNA expression for both genes was upregulated after HZE particle radiation exposure, with a significant decrease when cells were pretreated with SeM. We further investigated gene expression patterns of IL8 and TGFBeta after exposure to a high dose of HZE particle radiation (80 cGy), as well as a high dose of gamma radiation (4 Gy) by qPCR techniques. We observed a similar gene expression profile for these two genes, which was dependent on dose and type of radiation. Both IL8 and TGFBeta are closely associated with the carcinogenic process and with the radiation induced bystander effect. The fact that these genes are not induced by iron ion radiation exposure when cells are treated with SeM suggests potential beneficial effects for SeM. From these results, it is hypothesized that the ability of SeM to prevent HZE particle radiation induced cytotoxicity and/or transformation in vitro could be due to its ability to prevent, or reverse, the ability of iron ion radiation to induce IL8 and/or TGFBeta. C. 16 hour time point. In the past year, there has been further analysis of the differential expression of genes at 16 hours post-irradiation in cells irradiated with iron ions, in the presence or absence of SeM. At the earlier time points evaluated, there was extensive expression of genes participating in stress responses, but many of these gene expression changes were not observed at 16 hours post-irradiation. The SeM pretreated cells exposed to a 20 cGy iron ion radiation dose and processed 16 h post-irradiation, however, displayed down-regulation of genes associated with extracellular matrix remodeling (collagen 4, fibronectin 1 and laminins B1 and C1, and tenascin C) and cell-cell adhesion (N-cadherin). Altered levels of expression of these genes were not observed in irradiation alone conditions or at a 10 cGy dose with SeM pretreatment at either 6 or 16 h post-irradiation. Loss of normal tissue architecture and cell-extracellular matrix (ECM) interactions fundamentally drive tumor progression. These genes are known to have important roles in tumor growth, invasion, metastasis and angiogenesis. The fact that SeM leads to down-regulation of these genes at 16 hours post iron ion radiation exposure suggests that it may play an important role in inhibiting the expression of these genes associated with aggressive tumor phenotypes. Overall summary of gene expression studies. These results suggest that SeM may have numerous effects related to its ability to suppress HZE particle radiation induced transformation in vitro. From the studies performed as part of this grant, there are three major conclusions concerning the potential mechanism(s) for the anticarcinogenic effects of SeM, which are as follows: 1) SeM treatment results in a reduction in the levels of iron ion radiation induced IL8 and TGFBeta gene expression, which is expected to result in a suppression of carcinogenesis as well as an inhibition of radiation induced bystander effects, 2) SeM treatment of irradiated cells results in increased levels of p53 induction, which is expected to result in a suppression of carcinogenesis, and 3) SeM treatment results in reduced levels of the expression of genes associated with extracellular matrix remodeling (collagen 4, fibronectin 1 and laminins B1 and C1, and tenascin C) and cell-cell adhesion (N-cadherin) in cells exposed to iron ion irradiation, which is expected to result in a suppression of the progression of initiated cells to highly aggressive, malignant phenotypes.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Abstracts for Journals and Proceedings	Stewart JA, Ware JH, Fortina P, Breaux JK, Gulati S, Kennedy AR. "Gene expression in thyroid cells supplemented with L-selenomethionine - implications for cancer prevention in cells exposed to space radiation." Presented at the 8th International Conference on Nutrition and Cancer, Honolulu, Hawaii, March 10-12, 2005. Conference Proceedings, 8th International Conference on Nutrition and Cancer, pp. 33-34, March 2005. , Mar-2005
Abstracts for Journals and Proceedings	Zhou Z, Wan XS, Ware JH, Stewart JA, Kennedy AR. "The protective effects of L-selenomethionine on radiation induced malignant transformation." Presented at the 8th International Conference on Nutrition and Cancer, Honolulu, Hawaii, March 10-12, 2005. Conference Proceedings, 8th International Conference on Nutrition and Cancer, p. 57, March 2005. , Mar-2005
Abstracts for Journals and Proceedings	Kennedy AR, Ware JH, Fortina P, Breaux JK, Gulati S, Stewart JA. "Selenomethionine protects against space radiation biological effects." Presented at the 16th Annual NASA Space Radiation Investigators' workshop, Danfords on the Sound, Port Jefferson, New York, May 15-18, 2005. Conference Proceedings, 16th Annual NASA Space Radiation Investigators' Workshop, p. 106, May 2005. , May-2005

Abstracts for Journals and Proceedings	Stewart JA, Ware J, Fortina P, Kennedy AR. "Differential gene expression in thyroid epithelial cells exposed to low doses of HZE particle radiation." Presented at the NASA Bioinvestigators' Workshop, Galveston, Texas, January 10-12, 2005. Conference Proceedings, NASA Bioinvestigators' Workshop, p. 22, January 2005. , Jan-2005
Abstracts for Journals and Proceedings	Kennedy AR. "Prevention of space radiation carcinogenesis: Development of countermeasures for proton and HZE particle radiation induced transformation in vitro." Presented at the 11th International Charles Heidelberger Symposium on Cancer Research, Phitsanulok, Thailand, January 2006. Conference Proceedings, 11th International Charles Heidelberger Symposium on Cancer Research, pp. 28-29, January 2006. , Jan-2006
Abstracts for Journals and Proceedings	Kennedy AR, Ware JH, Fortina P, Breaux JK, Gulati S, Stewart JA. "Mechanisms by which L-selenomethionine protects against space radiation induced biological effects." Presented at the 4th International Workshop on Space Radiation Research and the 17th Annual NASA Space Radiation Health Investigators' Workshop, Moscow, Russia, June 7, 2006. Conference Proceedings, 4th International Workshop on Space Radiation Research and the 17th Annual NASA Space Radiation Health Investigators' Workshop, pp. 59-60, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Kennedy AR, Ware JH, Stewart JA. "Mechanisms by which selenomethionine protects against space radiation biological effects." Presented at the 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 6, 2006. Conference Proceedings, 53rd Annual Meeting of the Radiation Research Society, p. 68, November 2006. , Nov-2006
Abstracts for Journals and Proceedings	Kennedy AR, Stewart JA, Wan XS, Zhou Z, Ko Y-H, Wambi C, Ware JH. "Countermeasures for space radiation induced biological effects." Presented at the NASA Human Research Program Investigators' Workshop, League City, Texas, Feb. 13, 2007, Conference Proceedings, NASA Human Research Program Investigators' Workshop, p. 15, February 2007. , Feb-2007
Abstracts for Journals and Proceedings	Sanzari J, Ko Y-H, Kennedy AR. "Induction of IL8 in Htori-3 cells irradiated with HZE particles (iron ions)." Presented at the 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 14, 2007. Conference Proceedings, 18th Annual NASA Space Radiation Investigators' Workshop, p. 43, July 2007. , Jul-2007
Articles in Peer-reviewed Journals	Kennedy AR, Ware JH, Guan J, Donahue JJ, Biaglow JE, Zhou Z, Stewart J, Vazquez M,Wan XS. "Selenomethionine protects against adverse biological effects induced by space radiation." Free Radical Biology & Medicine 2004 Jan 15;36(2):259-66. PMID: 14744637 , Jan-2004
Articles in Peer-reviewed Journals	Stewart J, Ware J, Fortina P, Breaux J, Gulati S, Kennedy A. "L-selenomethionine modulates high LET radiation-induced alterations of gene expression in cultured human thyroid cells." Oncology Reports. 2006 Sep;16(3):569-74. PMID: 16865257 , Sep-2006
Articles in Peer-reviewed Journals	Kennedy AR, Zhou Z, Donahue JJ, Ware JH. "Protection against adverse biological effects induced by space radiation by the Bowman-Birk inhibitor and antioxidants." Radiation Research. 2006 Aug;166(2):327-32. PMID: 16881733 , Aug-2006
Articles in Peer-reviewed Journals	Stewart J, Ko Y-H, Kennedy AR. "Protective effects of L-selenomethionine on space radiation induced changes in gene expression." Radiation and Environmental Biophysics 2007 Jun;46(2):161-5. PMID: 17265150 , Jun-2007